

Jad Haddad

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EDUCATION

Texas A&M University

Bachelor of Science in Electronic Systems Engineering | 3.3 GPA

College Station, Texas

Aug. 2023 – Dec. 2027

Relevant Coursework: Power Systems & Circuits, Analog & Digital Electronics, Microcontroller Architecture, Embedded Systems Design

EXPERIENCE

Oncor Electric Delivery, Glen Rose Transmission

Protection & Control Engineering Intern

Glen Rose, Texas

May 2026 – Aug. 2026

- Programmed, configured, and field-tested protective relays across electromechanical, solid-state, and microprocessor-based systems at 345 kV and 138 kV transmission substations supporting Comanche Peak Nuclear Power Plant.
- Used Doble, Arbiter, and Omicron test sets for secondary current-injection and insulation-resistance testing to validate pickup thresholds, trip paths, and protection schemes against approved settings.
- Used HTML/SVG to build 4 interactive capacitor-bank simulations for a technical capstone presented to 25+ directors and senior engineers; adopted as the new company-wide standard training reference on capacitor banks.

ZKK Consulting LLC

Co-Founder, Technical & Patent Research Lead

Dallas-Fort Worth Metroplex, Texas

Aug. 2026 – Present

- Co-founded an engineering research firm focused on electronics, embedded systems, connected products, patents, components, and certification pathways, generating ~\$15,000 in the first month of operation.
- Conduct patent and prior-art research across medical devices, electronics, and hardware, mapping up to 15 patents per landscape to evaluate technical features, ownership, and status.

ENGINEERING PROJECTS

Vyrex, Wrist-Worn Fall-Detection Wearable

Founder, Hardware Lead

College Station, Texas

Feb. 2026 – May 2026

- Used KiCad to design a compact 4-layer mixed-signal PCB integrating an ESP32 MCU, 6-axis IMU, Li-Po charging, USB-C power, and MOSFET-driven haptics.
- Evaluated 3 MCUs for availability, power consumption, and pin compatibility, selecting a pre-certified module and avoiding ~\$17,000 in projected FCC intentional-radiator testing costs.
- Used an on-board Machine Learning Core and BLE connectivity to implement on-device fall detection with gait-based parameter adjustment through a custom companion app, achieving an 85% detection success rate in prototype testing.

Smart Table Solutions, Height-Adjustable Desk Control System

Founder, Self-Employed Electronics Engineer

Plano, Texas

May 2025 – Jul. 2025

- Designed and sold STM32-based motor-controlled table, generating ~\$2,400 across 8 units at a 100% on-time fulfillment rate.
- Engineered a 12 V H-bridge motor driver with overcurrent protection and closed-loop PWM control, validated in LTspice.

SoilSense, Automated Irrigation Controller

Personal Project

McKinney, Texas

Apr. 2025 – Jul. 2025

- Used C++ and an Arduino Uno R3 to automate irrigation with capacitive soil-moisture sensing, dual N-MOSFET pump control, and ultrasonic reservoir monitoring, achieving 270+ days of operation without intervention or failure.

INVOLVEMENT

Institute of Electrical and Electronics Engineers (IEEE)

Student Member

College Station, Texas

Dec. 2025 – Present

- Apply chapter workshop training in Keil μ Vision and KiCad to personal hardware projects for firmware bring-up and PCB layout.

Freshmen Exemplifying Aggie Spirit Together (FEAST)

Mentor & Former Special Events Planning Committee Member

College Station, Texas

Aug. 2023 – Present

- Mentor 15 undergraduate peers through their academic, organizational, and personal transition into Texas A&M
- Managed an ~\$8,000 semester budget and event coordination logistics for 96 peers on the Special Events Planning Committee.

Professional Association of Industrial Distribution (PAID)

Chapter Member

College Station, Texas

Dec. 2025 – Present

- Apply supplier network and supply-chain economics training from chapter workshops to component sourcing and BOM cost work.

TECHNICAL SKILLS

Languages: C, C++, Python, Java, ARM Assembly, Bash, MATLAB, Verilog, JavaScript, HTML/CSS, Tailwind

Hardware: Mixed-signal PCB design, Power Management & Li-Po Charging, IMU & Sensor Integration, BOM Development/Sourcing

Embedded: STM32, ESP32, BLE, UART, I2C, SPI, GPIO, PWM, ADC/DAC, ESP-IDF, Low-Power/Deep-Sleep Design, Edge Machine Learning

Tools: AcSElerator QuickSet, Powerbase, Multisim, Excel, Wireshark, STM32 Cube Programmer, Doble RTS, Linux, Arduino IDE

Certifications: Lean Six Sigma Yellow Belt (Texas A&M); Cloud Computing: Core Concepts (NASBA)